Assemble Your Toolbox

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Contents

1 Purpose

This document was created for the **Assemble Your Toolbox** course project in the Johns Hopkins University **Data Scientist's Toolbox** course on Coursera.

The GitHub repository for this project is available here:

GitHub Repository: JHU-Data-Scientists-Toolbox

Direct URL: https://github.com/treinart/JHU-Data-Scientists-Toolbox

The purpose of this project is to demonstrate that the basic tools used in the Data Science Specialization have been installed, configured, and connected successfully.

The tools demonstrated in this project are:

- R

- RStudio
- Git
- GitHub
- R Markdown
- CRAN packages
- Bioconductor packages

2 R and RStudio Setup

R and RStudio were installed locally and used to create this project. The project was managed through an RStudio Project folder and connected to GitHub using Git version control.

The local R version and working directory are shown below.

```r
R.version.string
```

```
## [1] "R version 4.5.2 (2025-10-31 ucrt)"
```

```r
getwd()
```

```
## [1] "C:/Users/travi/Documents/R-Programming-Course"
```

3 Package Setup

The following packages were installed during the course setup activities. The table below confirms that each package is available in my local R library.

```r
course_packages <- c(
  "ggplot2",
  "devtools",
  "lme4",
  "BiocManager",
  "GenomicFeatures",
  "rmarkdown",
  "knitr"
)

package_check <- data.frame(
  Package = course_packages,
  Installed = course_packages %in% rownames(installed.packages())
)

package_check
```

```
##           Package Installed
## 1         ggplot2      TRUE
## 2        devtools      TRUE
## 3            lme4      TRUE
## 4     BiocManager      TRUE
```

```
## 5 GenomicFeatures      TRUE
## 6       rmarkdown      TRUE
## 7           knitr      TRUE
```

The packages below are loaded to confirm that the installed R environment can run course-related package commands successfully.

```r
library(ggplot2)
library(lme4)
library(BiocManager)
library(GenomicFeatures)

cat("Course package check completed successfully.")
```

```
## Course package check completed successfully.
```

4 R Markdown Demonstration

R Markdown combines regular written explanation, R code, and code output in one reproducible document. This file was created in RStudio and knitted into a rendered output document.

The simple R code below confirms that code chunks run inside the R Markdown document.

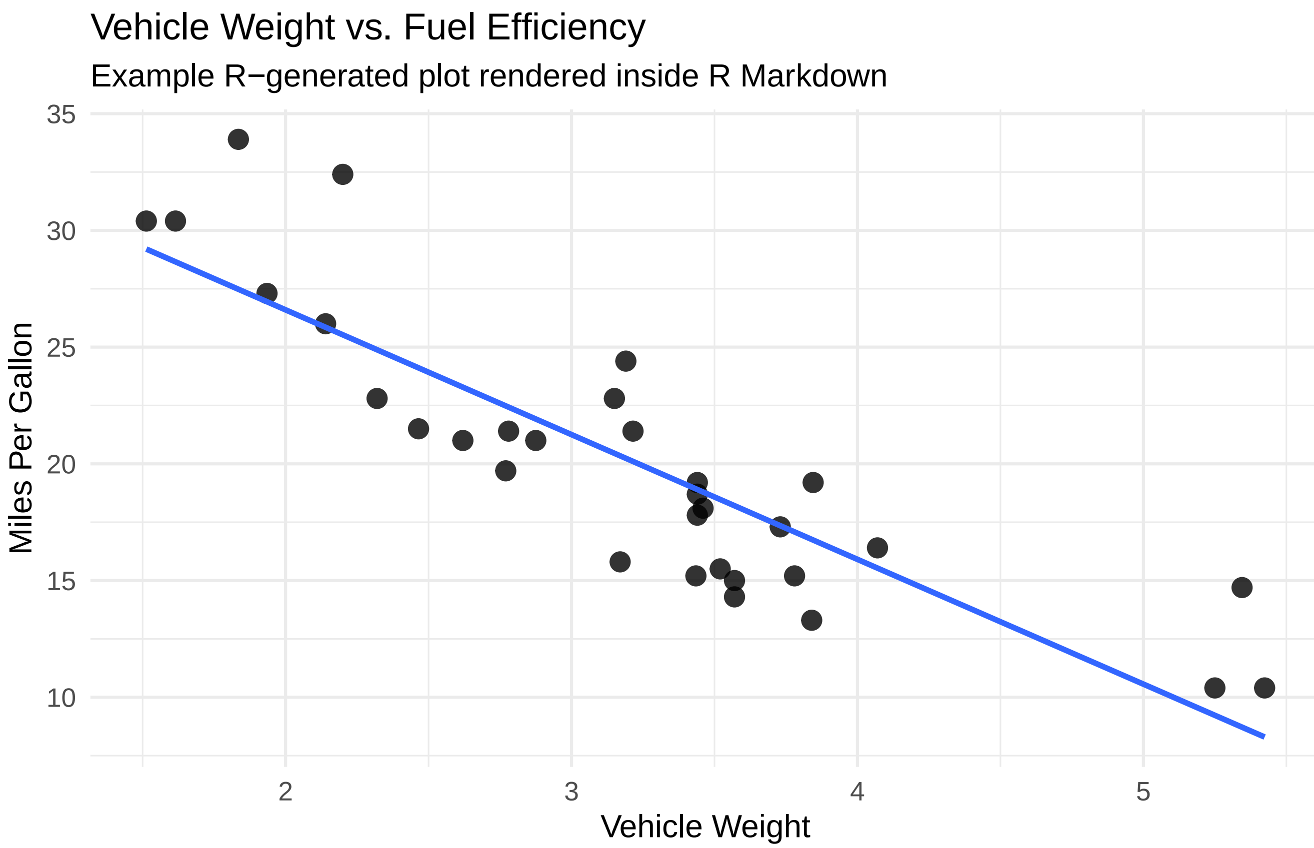
```r
x <- c(1, 2, 3, 4, 5)
mean(x)
```

```
## [1] 3
```

The plot below confirms that R Markdown can also include R-generated visual output.

```r
library(ggplot2)

ggplot(mtcars, aes(x = wt, y = mpg)) +
  geom_point(size = 3, alpha = 0.8) +
  geom_smooth(method = "lm", se = FALSE, linewidth = 1) +
  labs(
    title = "Vehicle Weight vs. Fuel Efficiency",
    subtitle = "Example R-generated plot rendered inside R Markdown",
    x = "Vehicle Weight",
    y = "Miles Per Gallon"
  ) +
  theme_minimal(base_size = 12)
```



5 Git and GitHub Setup

Git was used for local version control, and GitHub was used as the remote repository.

The workflow demonstrated in this project was:

1. Create and edit files in RStudio.
2. Stage files using Git.
3. Commit changes locally with a descriptive message.
4. Push committed changes to GitHub.
5. Confirm that the files appear in the GitHub repository.

The GitHub repository for this project is:

https://github.com/treinart/JHU-Data-Scientists-Toolbox

6 Setup Evidence

The screenshots below document the setup and version control workflow. The screenshots are stored in the `images/` folder of this repository and included here as supporting evidence.

6.1 RStudio Git Staging

This screenshot shows the RStudio Git pane with project files prepared for commit.

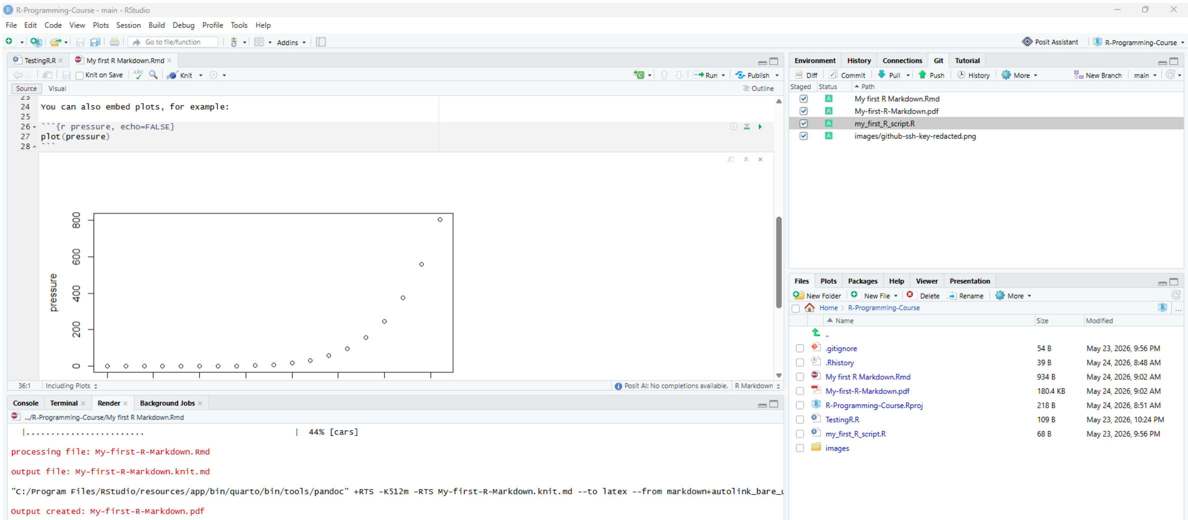


Figure 1: RStudio Git pane showing staged project files.

6.2 RStudio Review Changes and Commit Message

This screenshot shows the RStudio review window and the commit message used for the project files.

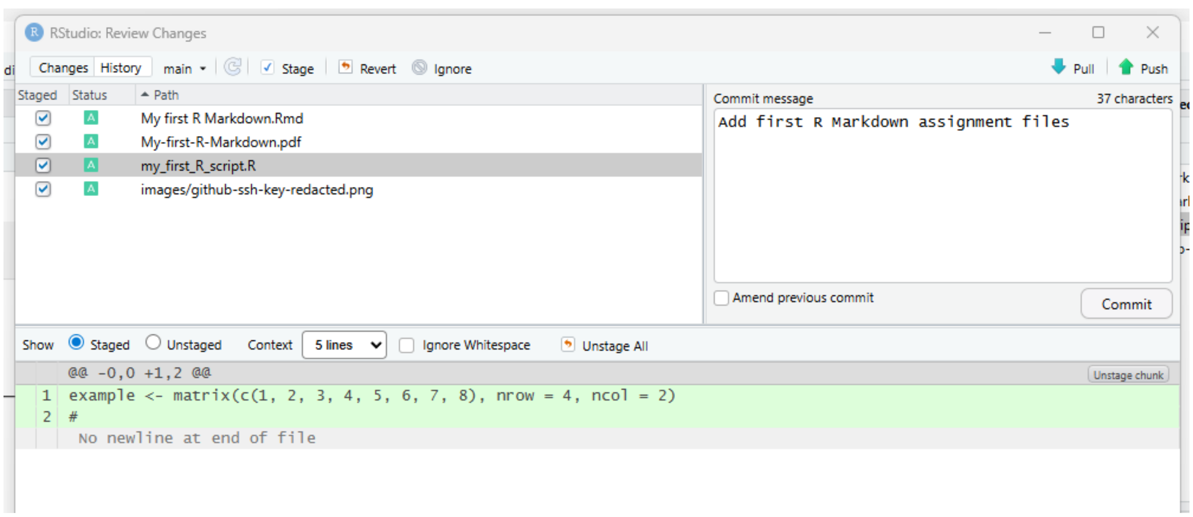


Figure 2: RStudio Review Changes window with commit message.

6.3 Git Push Confirmation

This screenshot shows the Git push confirmation from RStudio.

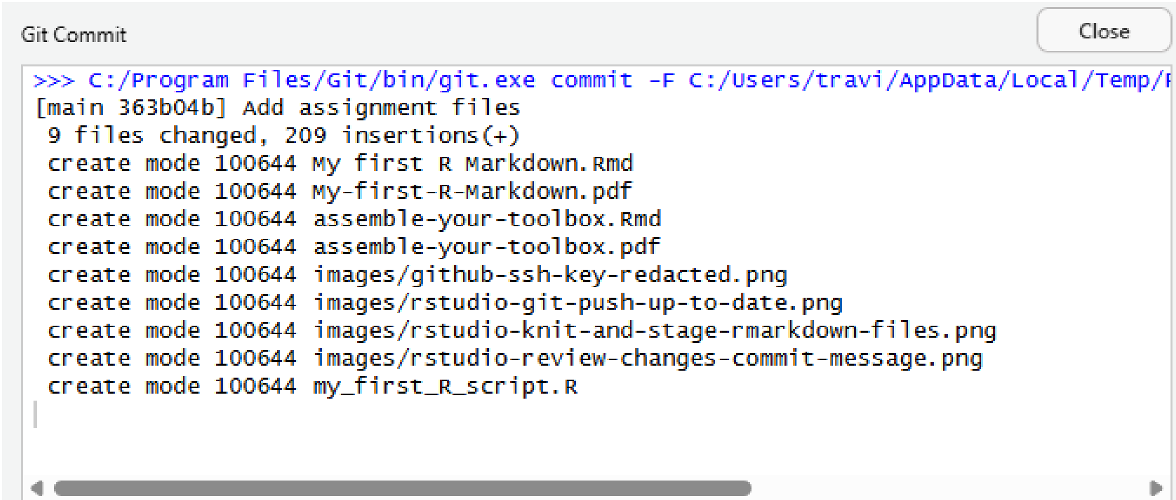


Figure 3: RStudio Git push confirmation.

6.4 GitHub SSH Key Setup

This screenshot shows the GitHub SSH key setup page with sensitive details redacted.

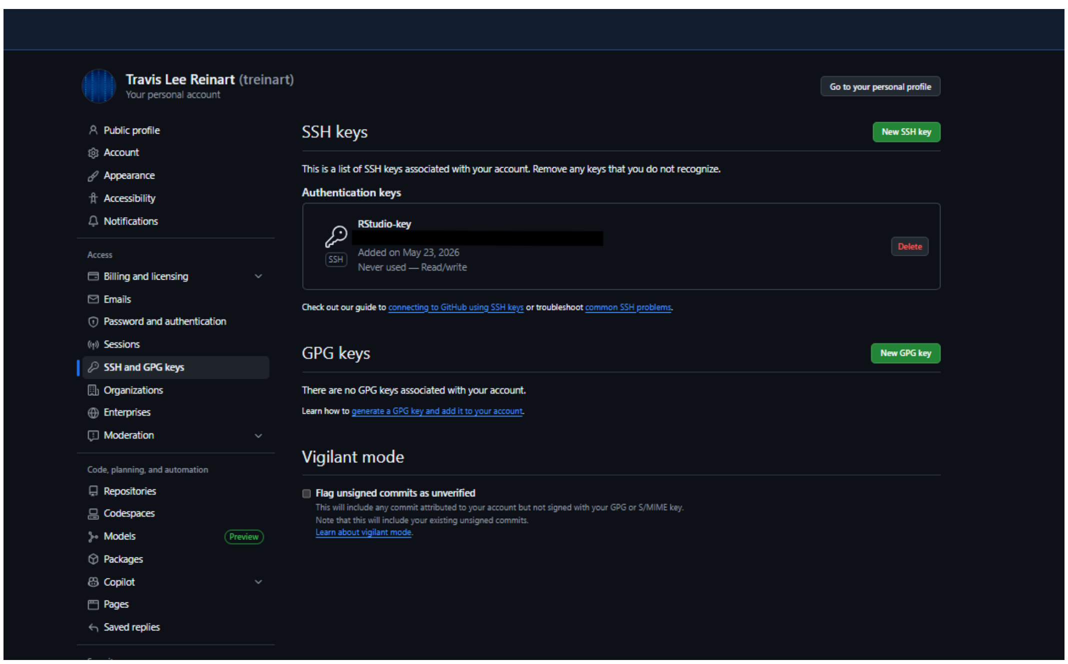


Figure 4: Redacted GitHub SSH key setup evidence.

6.5 GitHub Repository After Push

This screenshot shows the GitHub repository after the project files were pushed.

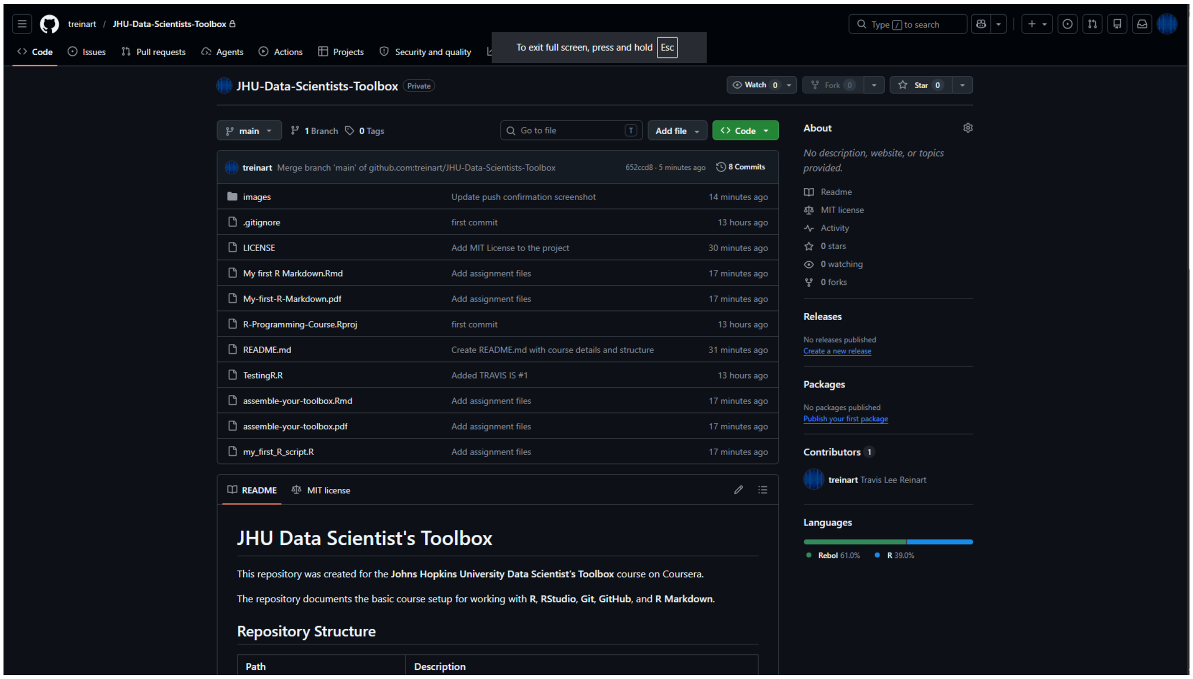


Figure 5: GitHub repository homepage after files were pushed.

7 Conclusion

This project demonstrates that the required course toolbox has been assembled. R and RStudio are installed locally, R packages can be installed and loaded, R Markdown can be created and rendered, Git is tracking the project files, and GitHub is receiving pushed commits from the local RStudio project.

The repository provides a working setup foundation for the rest of the Johns Hopkins Data Science Specialization.